

Buffered Viability and the Obstruction Margin

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Abstract

Exact viability is fragile: indicators carry measurement error, thresholds carry estimation error, and numerical methods need margins. This paper, the D12 item of the generalization catalogue, develops buffered viability: floors with a margin $\mathcal{V}_\delta = \{q_j \geq \delta \forall j\}$, and the *obstruction margin* $\mu(B)$ — the largest margin at which a belief remains obstructed — as the regulator’s measure of safety already lost to information constraints. On the two-patch working system the margin is computed exactly for the audited beliefs (e.g. $\mu(\{(1, 2), (2, 1)\}) = 2$ under both blind and full readings, maximality certified on the margin grid); viability is nested in the margin (viable at δ implies viable below), and the viable pair family erodes strictly under buffering ($28 \rightarrow 3 \rightarrow 0$ as $\delta = 0, 1, 2$). The timing layer’s buffered exit is exact twice: constant drift’s last safe step $\lfloor 10(z_0 - 1 - \delta) \rfloor$, and the contraction’s $\max\{k : q_0 \geq (\frac{10}{9})^k(1 + \delta)\}$ — the successor’s exit-time value with a multiplicative margin shift, verified at boundaries. The measurement reading is recorded: threshold mis-estimation up to $\mu(B)$ is covered exactly (6/6 check families).

Introduction and relation to the programme

The catalogue gives three reasons for buffering — measurement error, threshold uncertainty, numerical margins — and proposes the obstruction margin. The programme’s completed items supply the exact machinery whose buffered re-reading this paper performs: the structure verdicts of the successor, the exit-time values, the audited grids. No mechanism is changed; the floors acquire margins and everything is recomputed exactly.

The obstruction margin

For margin $\delta \geq 0$ let $\mathcal{V}_\delta$ be the buffered safe set and K_δ the observation-based kernel. The obstruction margin of a belief is

$$\mu(B) = \sup\{\delta : B \notin K_\delta\},$$

read as: how much safety margin the information constraint has already consumed.

Proposition 1 (Nesting and exactness). $\mathcal{V}_{\delta'} \subseteq \mathcal{V}_\delta$ for $\delta' \geq \delta$, hence viability at δ implies viability below; on the finite margin grid the supremum is attained, and maximality is certified

by the next grid point. For the audited belief $\{(1, 2), (2, 1)\}$: $\mu = 2$ under both the blind and full readings — the belief is obstructed even at the largest nontrivial margin, while singleton beliefs at full information retain positive margin.

Proof. The inclusion is definitional; obstruction transfers upward along the nested sets. Attainment on the grid is the maximum; the script certifies obstruction at μ and viability at the next margin with nonempty buffered set, for every reported value. $\square$

Proposition 2 (Buffered exit identities). With constant per-step decline $\frac{1}{10}$, the last safe step for the band $\{z \geq 1 + \delta\}$ equals $\lfloor 10(z_0 - 1 - \delta) \rfloor$ exactly. With contraction $\alpha(q) = q/10$, it equals $\max\{k : q_0 \geq (\frac{10}{9})^k(1 + \delta)\}$: the successor’s exit-time value shifts multiplicatively in the margin.

Proof. Arithmetic-progression and geometric-progression identities, exact in Fractions; the boundary equalities are part of the verified record. $\square$

Verified record (6/6 exact check families)

E1 the nesting chain over all 36 pairs and both audited structures; E2 the margins $\mu(\{(1, 2), (2, 1)\}) = \mu = 2$ (blind and full), $\mu(\{(2, 2)\}) = 2$ (full), each maximality certified; E3 the constant-drift identity on the 48-cell grid $\times$ margins $\{0, \frac{1}{10}, \frac{2}{10}\}$; E4 the contraction identity at boundaries and $\frac{1}{100}$ -neighbors ($K \leq 3$); E5 the strict erosion $28 \rightarrow 3 \rightarrow 0$; E6 the measurement reading: $V_2 \subseteq V_1 \subseteq V_0$ and obstruction at margin δ covers every true threshold at or below $1 + \delta$.

What is not claimed

No robust or chance-constrained viability theory is developed beyond the deterministic margin: the stochastic selector’s probabilities and the catalogue’s chance-constrained extension (its own “separate paper” warning is adopted) are not touched. The margin is computed on the finite grid, not estimated for continuous systems; no soft-constrained control design, no penalty methods. The erosion counts are instance facts, not a sensitivity theory. The obstruction margin measures information-caused safety loss on this system; generalizing the reading to institutional margins belongs to the decentralized companion.

Declarations

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Data availability. No external data were

used. **Code availability.** The verification script `buffered_viability_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all six check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.